5.4.8. Daughter recoil

If the β-particle and the neutrino are emitted with the same momentum but in opposite direction, the daughter nucleus experiences no recoil. On the other hand, if they are both emitted in the same direction, or if all the energy is carried away with one of the particles, the daughter experiences maximum recoil. The daughter therefore recoils with kinetic energies from zero up to a maximum value (when the β-particle is emitted with maximum energy). We can therefore write

$$Q_\beta = E_d + E_{max} \tag{5.43}$$

where E_d is the recoil energy of the daughter nucleus. From the laws of conservation of energy and momentum, and taking the relativistic mass changes of the electron into account, one finds that the daughter recoil energy is

$$E_d = \mathbf{m}_e E_{max}/m_d + E_{max}^2/\left(2\, m_d\, \mathbf{c}^2\right) \tag{5.44}$$

The recoil energy is usually ~100 eV, which still is sufficient for causing atomic rearrangements in surrounding molecules. In the decay of ^{14}C (to ^{14}N), E_{max} is 0.155 MeV, which gives $E_d = 7$ eV. However, by labelling ethane, $^{14}CH_3\,{}^{14}CH_3$, with ^{14}C in both C positions, it was found that $^{14}CH_3NH_2$ was formed in 50% of the cases when one of the ^{14}C atoms in ethane had decayed, although the C≡N bond strength is only 2.1 eV; most of the decays occur with less than the maximum recoil energy, which also can be averaged over the whole molecule. The small recoil also explains why decay reactions like

$$^{127}TeO_3{}^{2-} \rightarrow {}^{127}IO_3{}^{-} + \beta^- \tag{5.45}$$

and

$$^{52}MnO_4{}^{-} \rightarrow {}^{52}CrO_4{}^{2-} + \beta^+ \tag{5.46}$$

are possible, even when E_d is tens of electron volts. However, secondary effects tend to cause the chemical bond to break following radioactive decay (see §5.9).

5.5. GAMMA EMISSION AND INTERNAL CONVERSION

The α- and β-decay may leave the daughter nucleus in an excited state. This excitation energy is removed either by γ-ray emission or by a process called *internal conversion*.

5.5.1. Gamma emission

The α-emission spectrum of ^{212}Bi is shown in Figure 5.3. It is seen that the majority of the α-particles have an energy of 6.04 MeV, but a considerable fraction (~30%) of the α-particles have higher or lower energies. This can be understood if we assume that the

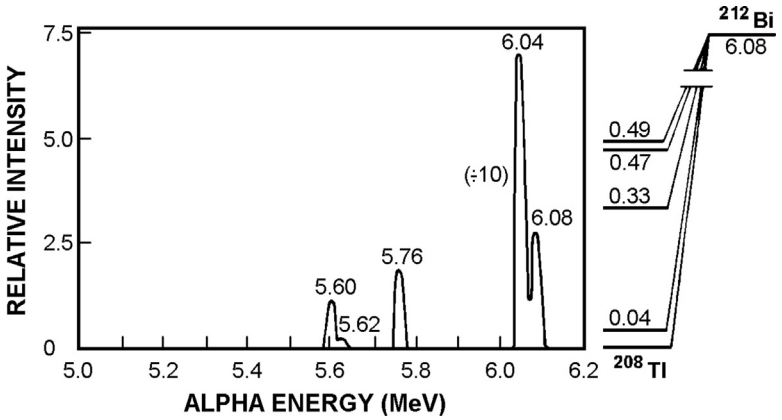


Figure 5.3 Alpha energy spectrum from ^{212}Bi → ^{208}Tl. (According to E. B. Paul.)

decay of parent ^{212}Bi leads to excited levels of daughter ^{208}Tl. This idea is supported by measurements showing the emission of γ-rays of energies which exactly correspond to the difference between the highest α-energy 6.08 MeV, and the lower ones. For example, an ~0.32 MeV γ accounts for the 5.76 MeV α (6.08 − 5.76 = 0.32). The excited levels of ^{208}Tl are indicated in the insert in Figure 5.3.

Gamma rays produce very low density ionization in gases so they are not usually counted by ionization, proportional, or Geiger counters. However, the fluorescence produced in crystals such as sodium iodide make scintillation counting of γ-rays efficient. Gamma ray spectra can be measured with very high precision using semiconductor detectors, cf §9.4. The higher the density of the material the more efficient the semiconductor material is to absorb the energy. Figure 5.4 shows such a spectrum for the decay of various excited states of ^{197}Au. The $7/2^+$, etc. symbols are explained in §5.8.

In the great majority of cases the emission of the γ-ray occurs immediately after α- or β-decay, i.e. within $\leq 10^{-12}$ s, but in some instances the nucleus may remain in the higher energy state for a measurable length of time. The longer-lived exited nuclei are called *isomers,* for definition of life time of isomers cf Ch. 1. An example is ^{60m}Co, which decays with a half-life of 10.5 min to the ground state of ^{60}Co. The decay is referred to as *isomeric transition*.

The decay energy in γ-emission is distributed between the γ-ray quantum (E_γ) and the kinetic energy of the recoiling product nucleus (E_d). We can therefore write

$$Q_\gamma = E_d + E_\gamma \tag{5.47}$$

The distribution of energy between the γ-ray and the recoiling daughter, according to

$$E_d = E_\gamma^2 / \left(2\, m_d\, \mathbf{c}^2\right) \tag{5.48}$$

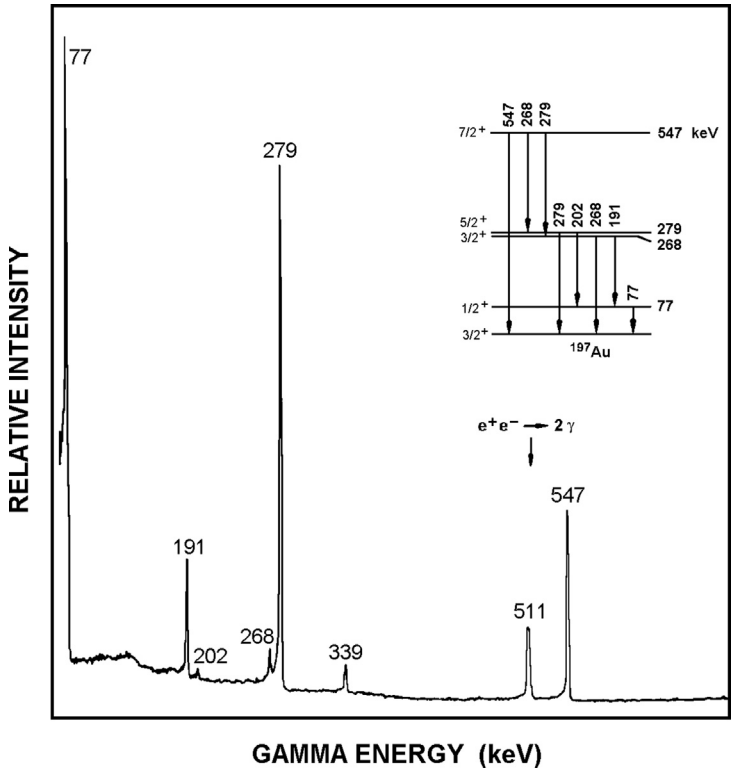


Figure 5.4 Gamma spectrum and decays scheme for ^{197}Au, produced through Coulomb excitation of gold by a 12 MeV ^{4}He beam. (According to M. G. Bowler.)

shows that $E_d < 0.1\%$ of E_γ. The amount of kinetic energy of the recoiling nuclide is therefore so trivial that it may be neglected when only the γ-ray energy is considered; cf. exercise 5.4. Even though in some cases the recoil energy from the gamma decay may be enough to break chemical bonds this is usually unimportant since the gamma follows a beta or alpha decay. Any bonds existing prior to the original decay are already been broken. However, for some longer lived meta stable nuclides, e.g. ^{99m}Tc new bonds could have been formed after the original decay and they may be broken by the gamma recoil.

5.5.2. Internal conversion

Gamma rays can interact with the orbital electrons of other atoms, so that the latter are expelled from that atom with a certain kinetic energy (see Ch. 7). A different process, called *internal conversion*, can occur *within* the atom undergoing radioactive decay. Because the wave function of an orbital electron may overlap that of the excited nucleus, the excitation energy of the nucleus can be transferred directly to the orbital electron

(without the involvement of a γ-ray), which escapes from the atom with a certain kinetic energy E_e. No γ-ray is emitted in an internal conversion process; it is an alternate mode to γ-ray emission of de-excitation of nuclei.

Internal conversion can be represented symbolically as

$$ {}^{Am}_{Z}X \rightarrow {}^{A}_{Z}X^{+} + {}^{0}_{-1}e^{-} \rightarrow {}^{A}_{Z}X \tag{5.49} $$

where we again have to consider the net atomic charge.

Part of the nuclear excitation energy is required to overcome the binding energy E_{be}, of the electron in its electronic orbital.[1] The remaining excitation energy is distributed between the recoiling daughter nucleus and the ejected electron E_e. The relationship is given by the equation

$$ Q_\gamma - E_{be} = E_d + E_e \tag{5.50} $$

The ejected electron, known as the *conversion electron*, normally originates from an inner orbital, since their wave functions have greater overlap with the nucleus. It is to be noted that the conversion electrons are mono-energetic, i.e. there is no neutrino emission associated with the ejected electron.

In as much as the binding energies of the atomic orbitals are different, the values of E_e reflect the differences in the electronic binding energies. In Figure 5.1 two sharp peaks are observed just beyond E_{max}. The first peak, designated as K, is due to conversion electrons originating in the K atomic shell, while the peak labeled L is due to conversion electrons originating in the L atomic shell. Both of these groups of conversion electrons arise from the decay of ^{137m}Ba. Figure 5.5(f) shows schematically the decay process of $^{137}Cs \rightarrow {}^{137}Ba + \beta^-$ for the principal decay path; IT is an abbreviation for isomeric transition. The decay of ^{137m}Ba proceeds both by emission of a 0.66 MeV γ-ray and by the competitive process of internal conversion. The ratio between the number of conversion electrons and the number of γ-rays emitted in this competition is called the *conversion coefficient*. The amount of internal conversion is not indicated in simplified decay schemes like Figure 5.5.

If we denote the conversion coefficient as α_i, it is equal to the ratio of K-electrons ejected (which we may denote with I_{eK}) to that of gamma quanta emitted (I_γ):

$$ \alpha_K = I_{eK}/I_\gamma \tag{5.51} $$

Usually $\alpha_K < 0.1$. Also $\alpha_K > \alpha_L > \alpha_M$, etc. For ^{137m}Ba the ratio of K-electrons to L-electrons emitted is about 5 while the value of α_K is 0.094.

It can be shown that the energy of the recoiling nucleus (E_d, (5.48)) is much smaller than the kinetic energy of the ejected electron E_e and may be ignored. The mathematical expression to use is (5.44).

[1] Electron-binding energies are tabulated in standard physics tables.

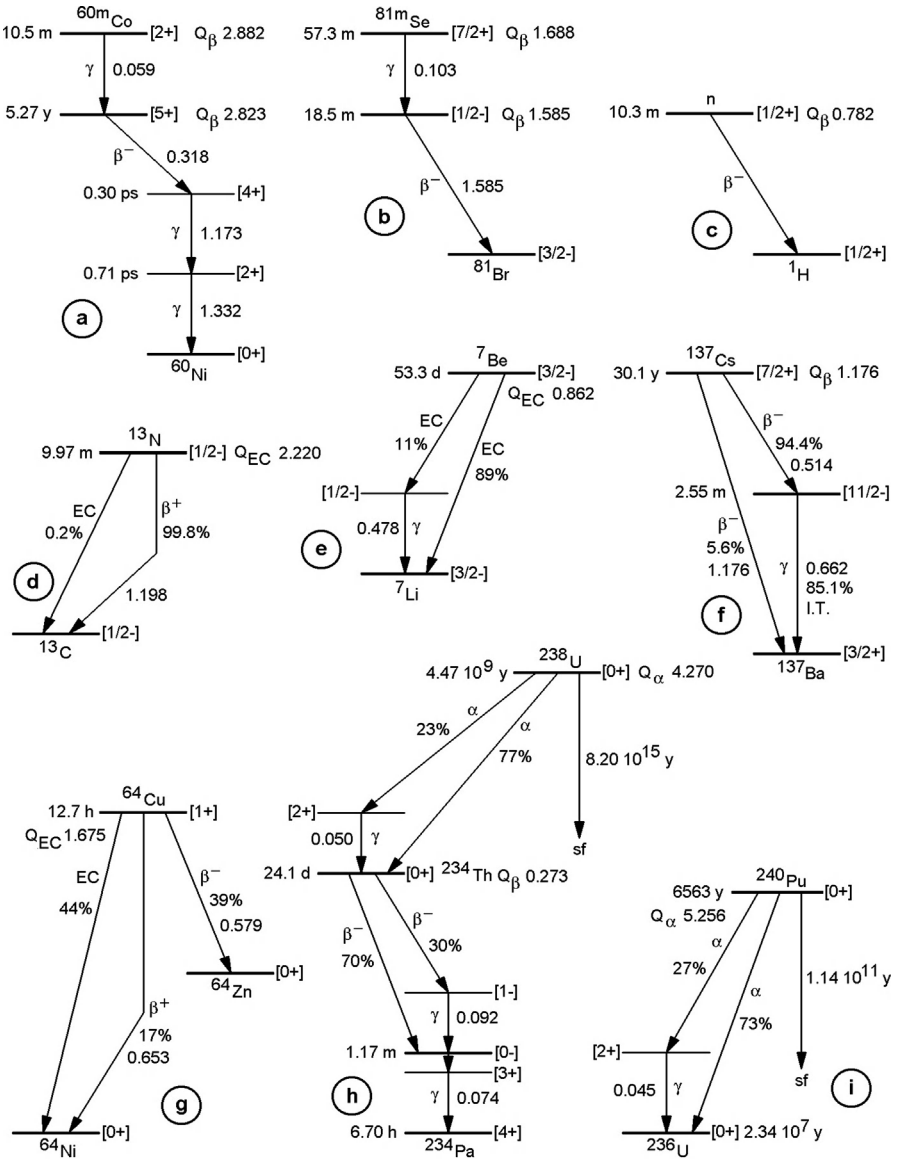


Figure 5.5 Examples of different decay schemes; energies are in MeV. The schemes are explained in the text.

A note on terminology. Consider Figure 5.5(f). Though the γ is emitted from an exited state of ^{137}Ba, in common language we nevertheless talks of "^{137}Cs-γ" (of 0.662 MeV), i.e. as if the γ was emitted from the ^{137}Cs nucleus. The quotation is to be interpreted as "^{137}Cs (decay through β^- followed by) γ(-emission)". A similar example is

from the decay of ^{60}Co where two cascade gamma photons are emitted from the excited ^{60}Ni nucleus. These two characteristic gamma lines of 1173 keV and 1332 keV are always referred to as "cobalt gammas" whereas they are actually emitted by nickel.

5.6. SPONTANEOUS FISSION

As the nuclear charge increases to large values, nuclei become more unstable. This is reflected by decreasing half-lives for nuclei heavier than uranium. In 1940 K. Petrzak and G. Flerov found that ^{238}U in addition to α-decay also had a competing mode of radioactive decay termed *spontaneous fission*. In this mode two heavy fragments (*fission products*) are formed in addition to some neutrons. The reaction may be written

$$ {}^{A}_{Z}\mathrm{X} \rightarrow {}^{A1}_{Z1}\mathrm{X}_1 + {}^{A2}_{Z2}\mathrm{X}_2 + \nu\mathrm{n} \tag{5.52} $$

where ν is the number of neutrons, usually 2–3. The half-life for spontaneous fission of ^{238}U is very long, about 8×10^{15} y. This means that about 70 fissions occur per second in 1 kg of ^{238}U, which can be compared with the simultaneous emission of 45×10^{9} α-particles.

With increasing Z, spontaneous fission becomes more common; i.e. the half-life for this decay process decreases. For $^{240}_{94}$Pu it is 1.2×10^{11} y; for $^{244}_{96}$Cm, 1.4×10^{7} y; for $^{252}_{98}$Cf, 66 y; and for $^{256}_{100}$Fm, 3×10^{-4} y. In fact, spontaneous fission becomes the dominating decay mode for the heaviest nuclei (see Fig. 4.3).

Spontaneous fission is in some ways similar to fission induced by bombardment with low energy neutrons (§11.7).

5.7. RARE MODES OF DECAY

Radioactive decay by *proton emission* is a very seldom observed decay mode for very neutron deficient nuclides because decay by β^+ or EC normally has a very much shorter partial half-life (§5.14). Decay by p^+ has been observed for ^{53m}Co (E_p 1.55 MeV, $t_{½}$ 0.25 s, ~1.5%). However, β^+ decay sometimes leads to a proton-unstable excited state, which immediately ($<10^{-12}$ s) emits a proton. Several β^+ emitters from ^{9}C to ^{41}Ti with $N = Z - 3$ have β^+ delayed proton emission with half-lives in the range 10^{-3}–0.5 s. Also radioactive decay by simultaneous emission of two protons has been observed for a few proton rich nuclides, e.g. ^{16}Ne, $t_{½}$ ~10^{-20} s.

Among the very neutron rich nuclides, e.g. some fission products, β^- delayed neutron emission is observed. This decay mode is similar in nature to the β^+ delayed p^+ emission. Delayed n-emission is important for the safe operation of nuclear reactors, see Ch. 19.

Decay by emission of particles heavier than α, e.g. ^{12}C, ^{16}O, is energetically possible for some heavy α-emitters and has been observed in a few cases.